



Workshop at the IOT Week
about

Globally Interoperable IoT Identification and Data Processing

6. June 2017, 9.00-17.00
International Conference Centre of Geneva
www.iot-week.eu

Whether in industry or science, data intensive projects are not at all efficient and cost too many resources excluding many smart people from participation in exploiting the treasure in data. With the current methods of managing and accessing data we are not fit to handle the continuous data streams with high granularity produced by the billions of smart devices being deployed in the coming years. We feel that it is time to change practices radically, but what are promising approaches to overcome the fragmentation?

Our current data practices are not fit to meet the data challenges of the coming decades!

Experts such as Robert Kahn will discuss about concepts that may open the doors for efficient data processing in the future.

Listen to distinguished experts from industry and science and discuss with them about their ideas. We are very much honoured to have one of the inventors of the Internet, Turing Award winner and designer of the Digital Object Architecture, Dr. Robert Kahn, in the group of experts to sharpen our understanding for the challenges and to identify possible solutions. The participation of experts from the worldwide active Research Data Alliance and International Telecom Union will also help to indicate directions.

Agenda

9.00	Peter Wittenburg (RDA)	Welcome and RDA Concepts for efficient data practices
9.30	Robert Kahn (CNRI)	From IP addresses to Persistent Identifiers – principles of Interlinking
10.15		Discussion
10.30	Coffee Break	
	Session 1 – Towards Interoperable Solutions	
11.00	Tobias Weigel (WDCC)	Infrastructure for the Global Climate Reporting

11.30	Ye Tian (CNIC-CAS)	Application of Digital Object Architecture in China
12.00	Peggy Irelan (Intel)	IoT Standards drive Frictionless Analytics
12.30	André Zwanziger (T-Systems)	MISP - Multi-IoT-Service-Plattform
13.00	Lunch Break	
	Session 2 – Towards Interoperable Solutions	
14.00	Alex Ntoko (ITU)	State of discussions in ITU
14.30	Jürgen Heiles (Siemens)	The role of Identifiers in IoT
15.00	Juanjo Hierro (FIWARE)	Supporting Researchers' Data Interoperability: FIWARE for the development of the IoT Cloud for RDA Europe
15.30	Coffee Break	
16.00	Leif Laaksonen (moderator): Robert Kahn, Tobias Weigel, Ye Tian, Peggy Irelan, André Zwanziger, Alex Ntoko, Jürgen Heiles, Juanjo Hierro, Peter Wittenburg	Expert Panel on Globally Interoperable IoT Identification and Data Processing – which ways to go?
17.00	End	

Panel Questions

Panel Objective: the aim of the panel discussion is to explore how industry and science can effectively collaborate on Globally Interoperable IoT Identification and Data Processing and what role RDA can play in facilitating this. At the end of the panel we would like to have a set of concrete actions that can be taken forward in the short-term future.

As a premise to the panel discussion, the following facts about data management and access are important to remember:

- about 60% of the efforts in data intensive projects in science and industry is wasted on data integration
- about 80% of scientists' time is wasted with simple data management tasks
- about 80% of the data generated is no longer available after a short period of time
- in science and industry we have a large number of infrastructure initiatives and cloud solutions serving urgent needs, but also creating an enormous fragmentation making data integration from different sources an extremely expensive task

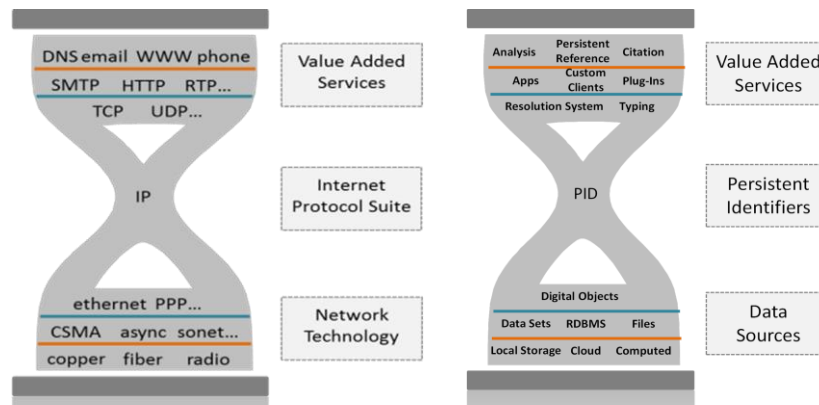


According to Intel and Oracle estimates, we will have 50 billion devices all creating continuous streams of highly granular data - orders of magnitude higher in terms of data volumes and complexity than what we currently have. We can simply state that we are not ready to efficiently deal with such data monsters and that we urgently need to change data practices and find global solutions.

This can lead us to a series of questions for the panel:

- For many of us it is evident that we need a change, but which kind of integration solutions do we currently see emerging and what is their potential?

- Expensive data integration from different sources forms a business model today. Is this model sustainable? How much will it dampen IoT business over time?
- What would be the gains compared to the losses for industry to adopt global interoperability standards?
- How far can we go with global agreement forming? Do we need to stick with an hourglass model as in the case of Internet (see the diagram in the attachment)? Is the hourglass still the best model?
- What is the role of the academia in globally interoperable IoT identification and data processing?



Speakers



Jürgen Heiles

Juergen Heiles has over 30 years of experience in Information and Communication Technologies from Satellite Communication to Optical Networking, IPTV, Smart Grid, IoT and Industrial Internet. He has been involved in standardization since the late 80s at ETSI, ITU, Open IPTV Forum, European Smart Grid Coordination Group and various other groups. Currently he oversees the Digitalization related standardization activities at Siemens Corporate Technologies. He is editor of IEEE P2413 Architecture Framework for Internet of Things and co-leader of the IoT Identifier task group in the Alliance for Internet of Things Innovations (AIOTI). He graduated from the University of Applied Sciences Koblenz, Germany.



André Zwanziger

André Zwanziger is Enterprise Architect and member of the Lead IoT Architects Team at Digital Division/T-Systems International GmbH (<http://www.t-systems.com>). T-Systems is a subsidiary of Deutsche Telekom AG with main focus on large enterprise customers. The Lead IoT Architects Team invented the award-winning Multi-IoT-Service-Platform in 2015/16 that is the fundamental base for IoT products of Deutsche Telekom and T-Systems (Experton I4.0/IoT Leader 2017 - <http://bit.ly/2oLJfZM>). Dr. André Zwanziger is author of articles in the

topics of DC automation, performance measurement for service management and IoT as well as a contributor for oneM2M (<http://www.onem2m.org/>). He is also part of the IoT strategy team at Deutsche Telekom.



Peggy Irelan (<https://www.linkedin.com/in/peggy-irelan-4138091>)

Peggy is an Intel Fellow, a Solution and Data architect, in the Data Center Solutions Group (DSG). Peggy provides technical leadership to the strategic DSG growth projects as well as lead the definition of data architecture, data models and technical governance model for DSG. Previous, Peggy was in Internet of Things Group working as a technology strategist and delivering end-to-end IoT analytics products in Intel® Decision Suite. This including sensors, software, and algorithms. These products turned the customers' data into trusted actionable intelligence to optimize their business. Previous to that, she was the chief architect in SSG, of Intel's Architecture for Monitoring Initiative. She led the initiative to create an innovative, reliable, and easy to use IA software and hardware platform monitoring and telemetry technology across all Intel product lines. This included the software framework solution that enables monitoring and telemetry driven Software & Services. Peggy started her career at Motorola where she developed and integrated MEMS sensor control systems for everything from cell phones, to run-flat tires, to air bags. Peggy has a B.S. in Computer and Electrical Engineering from Purdue University.



Juanjo Hierro (<https://www.linkedin.com/in/jhierro>)

Currently Chief Technology Officer (CTO) of the FIWARE Foundation. Also chairman of the FIWARE Technical Steering Committee. FIWARE is an Open Source initiative whose mission is to build an open sustainable ecosystem around public, royalty-free and implementation-driven software platform standards that ease the development of new Smart Applications in multiple sectors. The FIWARE platform combines components enabling the connection to the Internet of Things with Context Information Management and Big Data services on the Cloud. More info about FIWARE can be found at its website (<http://fiware.org>), YouTube channel (<http://www.youtube.com/user/FIWARE>) or following @FIWARE on Twitter.



Robert Kahn (https://en.wikipedia.org/wiki/Bob_Kahn)

Robert E. Kahn is Chairman, CEO and President of the Corporation for National Research Initiatives (CNRI), which he founded in 1986 after a thirteen year term at the U.S. Defense Advanced Research Projects Agency (DARPA). CNRI was created as a not-for-profit organization to provide leadership and funding for research and development of the National Information Infrastructure. Among many other awards, Bob Kahn was awarded the [SIGCOMM Award](#) in 1993 for "visionary technical contributions and leadership in the development of [information systems technology](#)", and shared the 2004 [Turing Award](#) with Vint Cerf, for "pioneering work on [internetworking](#), including the Internet's basic [communications protocols](#) and for inspired leadership in networking."



Leif Laaksonen (<http://www.leiflaaksonen.eu>)

Leif Laaksonen is currently a Director at CSC, the Finnish IT Center for Science located in Espoo, Finland, with responsibilities in international collaboration and project coordination/ management. His education is in quantum chemistry with a flavour of bioinformatics, from the late 80-ties and early 90-ties through practice. He has a long tradition in European research collaboration through chairing the e-Infrastructure Reflection Group (e-IRG) for the period 2007 – 2008 and 2009 – 2010 and serving as the e-IRG representative in the European Strategy Forum on Research Infrastructures (ESFRI) and as a e-IRG member in the ESFRI initiated Working Group on investment strategies in e-infrastructures.



Alexander Ntoko (<https://www.itu.int/en/ITU-T/Workshops-and-Seminars/20160628/Pages/NTOKO-Alexander.aspx>)

Alexander NTOKO is the Chief of the Operations and Planning Department in ITU where he plays a critical role in crafting and executing the strategic movements of ITU's Telecommunication Standardization Sector (ITU-T). He coordinates ITU activities on the Digital Object Architecture (DOA) and represents ITU in ICANN's Governmental Advisory Committee (GAC). Prior to taking up his current position, he was Head of ITU Corporate Strategy Division (CSD) where his responsibilities included ensuring organization-wide consensus and support for the development, implementation and evaluation of the ITU Strategic Plan. He assisted and advised the ITU Secretary-General in policies and strategies and was responsible for the coordination of ITU activities in many areas including Internet and cybersecurity. From the early 1990s, he played key role in the introduction of Internet and Cybersecurity to ITU. Since 1998, he has managed the implementation of projects on Internet, ICT applications (e.g., e-health, e-government, e-agriculture, e-education, e-payment and e-business) and cyber security including the use of advanced security technologies (biometric authentication and Public Key Infrastructure) in developing countries from all regions of the world. He obtained Bachelors (BSc) and Master of Science (MSc) degrees in Computer Science from the State University of New York.



Tobias Weigel (<https://www.dkrz.de/about/Organisation/mitarbeiter/TobiasWeigel>)

Tobias Weigel is a computer scientist working at the German Climate Computing Center (DKRZ) particularly in the areas of data infrastructures and persistent identifier services. Activities he is involved in include EUDAT services as well as big data management and analytics services for the Coupled Model Intercomparison Project phase 6 (CMIP6), which will generate the largest assembly of climate modelling data over the next years. Tobias holds a PhD in computer science from Hamburg University and is editorial board member of the CODATA Data Science Journal. Tobias was and is co-chair of

multiple working groups of the Research Data Alliance focused on PIDs, typing and actionable collections, and is a member of its Technical Advisory Board.



Peter Wittenburg (<http://www.mpcdf.mpg.de/~pewi>)

Peter Wittenburg was for many years technical director at the Max Planck Institute for Psycholinguistics (<http://www.mpi.nl>) and his group was responsible for technology and methodology development to facilitate the understanding how human brains process and acquire languages. Due to the outstanding work of his group he was appointed to become a member of the IT advisory board of the Max Planck Society. From about 2000 his group had leading roles in infrastructure building at national, European and international level (DOBES, CLARIN, EUDAT, ISOCAT). Due to his experience with the huge fragmentation he was one of the founders of the Research Data Alliance (<http://rd-alliance.org>). Recently he was invited by the German minister for education and science to lead a working group in the realm of the German IT Summit (<https://www.bildung-forschung.digital/de/plattform-digitalisierung-in-bildung-und-wissenschaft-1717.html>) and established intensive interactions with German IT industry.



Ye Tian (<http://www.escience.cn/people/tianyeiot/index.html>)

Tian is the executive deputy director of the Internet of Things Information Technology and Application Laboratory of the Computer Network Information Center (CNIC), Chinese Academy of Sciences (CAS). He is also the senior member of the China Computer Federation (CCF) and the liaison of the CCF Task Force on Big Data. Dr. Tian is responsible for the national information projects to ensure that every project can be carried out and be accomplished on time. Those projects include the National IoT Industrialization project, the China's Next Generation Internet project, The National Key Technology R&D Program, and so on. As the executive manager of the China National IoT Name Service Platform, he takes charge of the whole construction of the technology development and the nodes deployment.